

CA-C30L: Mobile Application Development Lab

LAB PROGRAM-8 : READ -WRITE LOCAL DATA

8. Create a program to Read/ write the Local data.

Steps to open and view the file after writing through the program

1. Select View->Tools Windows->device explorer(if using LDplayer,start the LD player before this)
2. Device explorer window will open
3. Select Data->data->[your package name]
->files->[your file name]

Note : If AAPT Compiler parser error is encountered :

1. when you get the error, at the bottom of Android Studio click the Build tab (assuming it's not already open). Once the Build tab opens, in the left pane **click** :app:mergeDebugResources (green arrow in pic). **Look closely and it says 1 error**
2. After you click **:app:mergeDebugResources**, then in the right pane, scroll to the top and you are going to see a blue link with a red error message next to it (underlined in green in pic). The red error message describes what's causing the issue. For eg. the issue could be **Invalid >color< for given resource value**. Click the blue link and it'll take you to the file that contains the issue.

Make changes in the java and xml file as follows :

Mainactivity.java

```
package com.example.readwritenew;

import android.content.Context;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```

import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;

public class MainActivity extends
AppCompatActivity implements View.OnClickListener
{

    // declare the variables
    Button read, write;
    EditText userInput;
    TextView fileContent;
    String filename = "myfile.txt"; // Define
your filename here

    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Handle system bars (optional)
        /*

ViewCompat.setOnApplyWindowInsetsListener(findView
ById(R.id.main), (v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBar
s());

            v.setPadding(systemBars.left,
systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });*/

        // Initialize the UI components
        read = findViewById(R.id.read_button);
        write = findViewById(R.id.write_button);

```

```

        userInput = findViewById(R.id.userInput);
        fileContent = findViewById(R.id.content);

        // Set click listeners
        read.setOnClickListener(this);
        write.setOnClickListener(this);
    }

    @Override
    public void onClick(View view) {
        Button b = (Button) view;

        // Get the button text: either 'read' or
        'write' depending on the button pressed
        String b_text = b.getText().toString();

        switch (b_text.toLowerCase()) {
            case "write": {
                writeData();
                break;
            }
            case "read": {
                readData();
                break;
            }
        }
    }

    private void writeData() {
        try {
            // Write the input text to the file
            FileOutputStream fos =
openFileOutput(filename, Context.MODE_APPEND);
            String data =
userInput.getText().toString();
            fos.write(data.getBytes());
            fos.flush();
            fos.close();

```

```

        } catch (IOException e) {
            e.printStackTrace();
        }

        userInput.setText(""); // Clear the input
        field
        printMessage("Writing to file " + filename
+ " completed...");
    }

    private void readData() {
        try {
            // Read the content of the file
            FileInputStream fin =
openFileInput(filename);
            int a;
            StringBuilder temp = new
StringBuilder();
            while ((a = fin.read()) != -1) {
                temp.append((char) a);
            }

            // Set the file content to the
TextView
            fileContent.setText(temp.toString());
            fin.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
        printMessage("Reading from file " +
filename + " completed...");
    }

    // A simple method to print messages to the
fileContent TextView
    private void printMessage(String message) {
        fileContent.append("\n" + message);
    }

```

```
}  
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>  
<androidx.constraintlayout.widget.ConstraintLayout  
out  
xmlns:android="http://schemas.android.com/apk/res/  
android"  
  
xmlns:app="http://schemas.android.com/apk/res-  
auto"  
  
xmlns:tools="http://schemas.android.com/tools"  
    android:id="@+id/main"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    tools:context=".MainActivity">  
  
    <TextView  
        android:id="@+id/textView2"  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:text=" File Content "  
        android:textAlignment="center"  
        android:textColor="#000"  
        android:textSize="24sp"  
        android:textStyle="bold"  
  
        app:layout_constraintBottom_toBottomOf="parent"  
  
        app:layout_constraintLeft_toLeftOf="parent"  
  
        app:layout_constraintRight_toRightOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent"

app:layout_constraintVertical_bias="0.52" />
```

```
<Button
    android:id="@+id/write_button"
    android:layout_width="wrap_content"
    android:layout_height="48sp"
    android:layout_marginStart="160sp"
    android:layout_marginEnd="159sp"
    android:layout_marginBottom="16sp"
    android:text="Write"
```

```
app:layout_constraintBottom_toTopOf="@+id/read_
button"
```

```
app:layout_constraintEnd_toEndOf="parent"
```

```
app:layout_constraintHorizontal_bias="0.0"
```

```
app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent"
```

```
app:layout_constraintVertical_bias="0.904" />
```

```
<Button
    android:id="@+id/read_button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="160sp"
    android:layout_marginEnd="158sp"
    android:layout_marginBottom="48sp"
    android:text="Read"
```

```
app:layout_constraintBottom_toTopOf="@+id/textV  
iew2"
```

```
app:layout_constraintEnd_toEndOf="parent"
```

```
app:layout_constraintHorizontal_bias="0.0"
```

```
app:layout_constraintStart_toStartOf="parent"  
/>
```

```
<EditText
```

```
    android:id="@+id/userInput"  
    android:layout_width="wrap_content"  
    android:layout_height="50dp"  
    android:layout_marginStart="46sp"  
    android:layout_marginTop="91sp"  
    android:layout_marginEnd="46sp"  
    android:ems="10"  
    android:inputType="textPersonName"  
    android:text="Name"
```

```
app:layout_constraintEnd_toEndOf="parent"
```

```
app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent"  
    android:hint="enter"/>
```

```
<TextView
```

```
    android:id="@+id/content"  
    android:layout_width="332sp"  
    android:layout_height="306sp"  
    android:layout_marginStart="33sp"  
    android:layout_marginTop="21sp"  
    android:layout_marginEnd="33sp"
```

```
        android:layout_marginBottom="6sp"
        android:text=""
        android:textAlignment="center"
        android:textColor="#000000"

app:layout_constraintBottom_toBottomOf="parent"

app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.461"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/textV
iew2"

app:layout_constraintVertical_bias="0.0" />
</androidx.constraintlayout.widget.ConstraintLa
yout>
```
